SQL LAST-DAY INTERVIEW CHEATSHEET

SELECT BASICS

SELECT * FROM table;
SELECT col1, col2 FROM table;
SELECT DISTINCT col FROM table;

WHERE vs HAVING

WHERE filters rows (before GROUP BY).
HAVING filters groups (after GROUP BY).

AGGREGATE FUNCTIONS

COUNT, SUM, AVG, MAX, MIN

GROUP BY RULE

Every column in SELECT must be aggregated or included in GROUP BY.

JOINS

INNER JOIN → matching rows
LEFT JOIN → all left rows

SELF JOIN

Used for hierarchical data like employee-manager relationships.

SUBQUERIES

Salary greater than average, second highest salary.

DUPLICATES

GROUP BY + HAVING COUNT(*) > 1 to find duplicates.

UPDATE

UPDATE table SET col = value WHERE condition;

DELETE (SAFE PRACTICE)

Always run SELECT before DELETE.

ALTER

ADD, MODIFY, DROP, RENAME columns.

CONSTRAINTS

PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL.

NULL HANDLING

Use IS NULL / IS NOT NULL.

LIKE PATTERNS

A% (starts), %A (ends), %A% (contains).

WINDOW FUNCTIONS

ROW_NUMBER, RANK, DENSE_RANK.

INTERVIEW ONE-LINERS

WHERE filters rows, HAVING filters groups.
Indexes speed reads but slow writes.